A Rank and Signature Obstruction to Lorentzian Metric Emergence from Positive-Sign Scalar Gradients

Richard J. Reyes

October 26, 2026

Abstract

We establish a pointwise algebraic obstruction to constructing a four-dimensional Lorentzian metric from the real part of the derivative product of a single complex scalar field. Writing the scalar as two real components shows that its proposed symmetric tensor is a Gram form with rank at most two and is positive semidefinite. Consequently, the tensor is degenerate and cannot have Lorentzian signature. More generally, any finite sum of real scalar-gradient outer products with nonnegative weights is positive semidefinite, regardless of how many gradient directions are supplied. The conclusion is specific to this positive-sign gradient construction; it does not exclude composite or effective metrics involving different tensorial ingredients or independent geometric structure.

1 Setup and main theorem

Let M be a smooth real four-dimensional manifold and let $\psi \in C^1(M;\mathbb{C})$ be a complex scalar field. Write

$$\psi = a + ib, \qquad a, b \in C^1(M;\mathbb{R}). \tag{1}$$

Define the real symmetric covariant $(0,2)$-tensor

$$T_{\mu\nu} := \mathrm{Re}(\partial_\mu \overline{\psi}\, \partial_\nu \psi). \tag{2}$$

This definition is coordinate independent because derivatives of scalar fields are covectors.

Theorem 1 (Single-complex-scalar rank and signature obstruction)**.** *At each point $p \in M$, the bilinear form T_p defined in* (2) *satisfies*

$$\operatorname{rank} T_p \le 2, \tag{3}$$

$$T_p(V,V) \ge 0 \quad \textit{for every } V \in T_pM, \tag{4}$$

$$\dim \ker T_p \ge 2. \tag{5}$$

Here $\ker T_p = \{V \in T_pM : T_p(V,W) = 0 \textit{ for all } W \in T_pM\}$ *is the radical of the bilinear form. In particular,* $\det T_p = 0$ *in any coordinate basis, and T_p cannot be a Lorentzian metric.*

Proof. Since $\overline{\psi} = a - ib$,

$$\partial_\mu \overline{\psi}\, \partial_\nu \psi = (\partial_\mu a - i\partial_\mu b)(\partial_\nu a + i\partial_\nu b) \tag{6}$$

$$= \partial_\mu a\, \partial_\nu a + \partial_\mu b\, \partial_\nu b + i(\partial_\mu a\, \partial_\nu b - \partial_\mu b\, \partial_\nu a). \tag{7}$$

Taking the real part gives the exact identity

$$T_{\mu\nu} = \partial_\mu a\, \partial_\nu a + \partial_\mu b\, \partial_\nu b. \tag{8}$$

Fix $p \in M$ and a coordinate basis for T_pM. Set $A_\mu = (\partial_\mu a)_p$ and $B_\mu = (\partial_\mu b)_p$. Introduce the real 4×2 matrix

$$J = \begin{pmatrix} A_0 & B_0 \\ A_1 & B_1 \\ A_2 & B_2 \\ A_3 & B_3 \end{pmatrix}. \tag{9}$$

Equation (8) is the Gram factorization

$$[T_p] = JJ^{\mathsf{T}}. \tag{10}$$

It follows that

$$\operatorname{rank} T_p = \operatorname{rank}(JJ^{\mathsf{T}}) \le \operatorname{rank} J \le 2, \tag{11}$$

proving (3). In particular, all 3×3 minors of $[T_p]$ vanish, and its 4×4 determinant vanishes.

For any real tangent vector $V = V^\mu \partial_\mu|_p$,

$$T_p(V,V) = (V^\mu \partial_\mu a)^2 + (V^\mu \partial_\mu b)^2 \tag{12}$$

$$= \|J^{\mathsf{T}}[V]\|^2_{\mathbb{R}^2} \ge 0. \tag{13}$$

This proves positive semidefiniteness. Finally, rank–nullity gives

$$\dim\ker T_p = 4 - \operatorname{rank} T_p \ge 2. \tag{14}$$

The kernel consists precisely of the tangent vectors annihilated by both $\mathrm{d}a|_p$ and $\mathrm{d}b|_p$, as follows from (12) and (8).

A Lorentzian metric in four dimensions is a nondegenerate symmetric bilinear form of inertia $(1,3)$ or $(3,1)$: it is indefinite and has rank four. By contrast, T_p is both degenerate and positive semidefinite. Either property alone excludes Lorentzian signature. These properties are invariant under changes of coordinates (the inertia statement follows from Sylvester's law of inertia). Therefore T_p cannot be a Lorentzian metric. □

2 Positive-sign multiplets

Proposition 1 (Nonnegative scalar-gradient sums)**.** *Let $\phi^1, \dots, \phi^N \in C^1(M;\mathbb{R})$ and let $w_A : M \to [0,\infty)$ be nonnegative weights. Define*

$$S_{\mu\nu} := \sum_{A=1}^{N} w_A\, \partial_\mu \phi^A\, \partial_\nu \phi^A. \tag{15}$$

At every $p \in M$,

$$\operatorname{rank} S_p \le \min\{N, 4\}, \qquad S_p(V,V) \ge 0 \quad \textit{for all } V \in T_pM. \tag{16}$$

Thus no such tensor is Lorentzian, for any finite N.

Proof. At a fixed p, define the real $4 \times N$ matrix

$$K_{\mu A} := \sqrt{w_A(p)}\,(\partial_\mu \phi^A)_p. \tag{17}$$

Then $[S_p] = KK^{\mathsf{T}}$, and hence

$$\operatorname{rank} S_p \le \operatorname{rank} K \le \min\{N, 4\}. \tag{18}$$

Moreover, for every real $V \in T_pM$,

$$S_p(V, V) = \sum_{A=1}^{N} w_A(p)\big(V^\mu \partial_\mu \phi^A\big)^2 \ge 0. \tag{19}$$

A Lorentzian form is indefinite; therefore S_p cannot be Lorentzian, even if it has rank four. □

Corollary 1 (Rank budget is necessary but not sufficient)**.** *Within the construction* (15)*, at least four independent real gradient directions are necessary to obtain a rank-four tensor at a point. Having four or more such directions cannot by itself produce Lorentzian signature when the weights remain nonnegative.*

3 Sharp examples and the role of additional structure

Example 1 (Sharpness of the rank-two bound). In coordinates (x^0, x^1, x^2, x^3) on $\mathbb{R}^4$, choose $a = x^1$ and $b = x^2$. Then

$$[T] = \operatorname{diag}(0, 1, 1, 0), \tag{20}$$

so $\operatorname{rank} T = 2$ exactly.

Example 2 (Four gradients give full rank, not Lorentzian signature). Take $\phi^A = x^{A-1}$ for $A = 1, 2, 3, 4$ and $w_A = 1$. Then

$$[S] = \operatorname{diag}(1, 1, 1, 1), \tag{21}$$

which is full rank but positive definite, not Lorentzian.

Example 3 (Indefinite internal bilinear form). Suppose instead one introduces an internal tensor $\eta_{AB} = \operatorname{diag}(-1, 1, 1, 1)$ and defines

$$G_{\mu\nu} := \eta_{AB}\,\partial_\mu \phi^A\,\partial_\nu \phi^B. \tag{22}$$

For $\phi^A = x^A$ with indices $A = 0, 1, 2, 3$, one gets

$$[G] = \operatorname{diag}(-1, 1, 1, 1). \tag{23}$$

Thus Lorentzian signature is algebraically possible once an indefinite internal bilinear form is supplied. The internal form is additional input, not a consequence of (2).

Remark 1 (Scope of the no-go statement). The theorem excludes identifying the specific tensor $\operatorname{Re}(\mathrm{d}\overline{\psi} \otimes \mathrm{d}\psi)$, or any nonnegative scalar-gradient sum of the form (15), with a Lorentzian metric. It does not prove that no theory involving a complex scalar can possess an effective Lorentzian metric. In particular, it does not address constructions using an independent metric, indefinite contributions, disformal combinations, additional tensor fields, or other geometric data. Nor does it establish Einstein dynamics, diffeomorphism gauge structure, a massless spin-two mode, or a microscopic theory of gravity.

4 Conclusion

The proposed tensor is a Gram form of only two real covectors:

$$\boxed{T_{\mu\nu} = \partial_\mu a\, \partial_\nu a + \partial_\mu b\, \partial_\nu b, \quad \operatorname{rank} T \leq 2, \quad T(V,V) \geq 0.} \tag{24}$$

Thus it is everywhere degenerate and positive semidefinite, and cannot be a four-dimensional Lorentzian metric. More positive-sign scalar gradients can remove the rank deficiency, but cannot change the positive-semidefinite signature. Lorentzian metric emergence therefore requires a different tensor construction, an indefinite contribution, or additional geometric structure.

Research context. This observation is applied to the tensor/tetrad emergence question in the foundational audit of the geometry-of-resonance program: `https://github.com/rickyjreyes/geometry_of_resonance/blob/main/docs/FOUNDATIONAL_STAGE2_2026-10-14.md`. The underlying rank and Gram-matrix facts are standard linear algebra; the no-go conclusion concerns the specified candidate construction.